MVP validation

Evidence draft - November 5, 2026. Reviewed production data and independent local numerical/UI validation are complete. Public screenshots, hosted access, user-owned interpretations and video are pending. This PDF does not claim public-app verification.

This dashboard helps a CPSC epidemiology analyst prioritize questions about treadmill injury mechanisms among adults 65-120. It combines weighted ED burden, recorded severe outcomes, primary diagnoses and original narratives. The genuine 2025 cohort contains 109 approved visits from 188 screened candidates; 79 are excluded under Ryan's accepted injury-focused policy. The annual source has 410,201 records and 25 original fields. One row is one ED visit; no exposure-adjusted risk, causation or automatic priority is inferred.

1. Design

The CPSC NEISS FAQ describes surveillance estimates and follow-back investigation, and distinguishes product association from causation. This supports an investigation-screening use case. Our particular interface, role scenarios and prioritization workflow are design hypotheses; no interview, usability study or observed analyst adoption is claimed. PENDING: team/user corroboration of decision constraints, workflow and usefulness.

The intended decision is where to spend limited follow-up attention, rather than whether a product is defective. Burden and severe share answer different questions: estimated volume versus recorded outcomes among visits. Diagnosis and narratives add context; visible evidence and missing outcomes help prevent overreading a high percentage supported by few records. The following eight changes extend the professor's starting build and remain protected by automated tests.

Change request	Implemented behavior and decision rationale
CR-01	Selected severe burden alongside all-outcome severe share, retaining mechanism/search. Prevents a severe-only filter from flattening every nonempty percentage to 100%.
CR-02	Visible mechanism record counts and unknown-disposition counts/weights. Makes evidence volume and missingness inspectable without inventing precision thresholds.
CR-03	Primary-diagnosis baseline beside selected composition under severity filtering. Keeps clinical context from being silently redefined.
CR-04	Matching IDs and original narratives after all active filters. Enables users to examine what a literal search actually selected.
CR-05	Reset and session-specific Undo reset. Restores the complete prior mechanism/severity/search selection for comparison.
CR-06	Unknown share and none/all-unknown-severe endpoints. Shows the decision impact of missing outcomes without imputing values.
CR-07	In-app mechanism definitions, version and initiating-event precedence. Makes taxonomy interpretation visible.
CR-08	Full-precision contribution reconciliation and CSV with filter/period/data hash. Explains display rounding and supports independent audit.

The team deliberately did not build exposure-adjusted risk, geographic/brand attribution, prediction, causal defect detection, significance claims or automatic investigation recommendations. The data and design do not support them.

Source period is fixed to 2025; year-range controls and trends would require additional agreed artifacts and sampling interpretation. Counts are not confidence intervals, and no design-based uncertainty is estimated.

Meaningful alternatives are the official NEISS query builder, an independent spreadsheet/script, and CPSC follow-back investigation. The official query provides selectable confidence intervals/CV and downloadable records; it does not automatically reproduce our narrative-screened, manually classified cohort. A separate analysis can explore broader fall-involving definitions but takes reconciliation work. Follow-back can supply circumstances absent from short narratives; this dashboard cannot replace it. A database or automated classifier does not supply missing narrative circumstances or evidence of usefulness. Small, precomputed flat files keep interactions in memory and avoid exposing preparation inputs.

Preparation, loading, filtering, aggregation and UI wiring remain separate. Source and candidate fingerprints bind a completed review to its exact export; the loader rejects missing, edited, incomplete or sample artifacts instead of silently displaying national estimates. The injury-focused policy narrows the original broader involvement contract. Ryan explicitly accepts the matching policy and confirms his case review. The policy document retains its original authorship/fingerprint, and the actual acceptance is preserved in private review documentation. His subsequent version 1.2 taxonomy clarification supersedes only the old mechanism-version reference; inclusion scope and the original prompt are not silently changed. All records and taxonomy boundaries require real human verification; AI agreement does not satisfy that gate.

PENDING LIVE EVIDENCE: initial public screenshot with period, labels and selections; user/team assessment of whether these features support the intended decision. The code and tests are evidence of implemented behavior, not of actual user benefit.

The accepted 1.2 clarification replaces a use-status assumption with Equipment/cord obstacle or landing falls. It describes eight equipment-relationship cases without asserting non-use, motor state or the initial cause of imbalance. Operating/on-from-supporting-surface falls remain Fall; more specific initiating mechanisms retain precedence. The four final mappings were explicitly authorized in Ryan's FINAL RELEASE message. The original 171 approvals and their later personal confirmation are preserved; no additional independent inspection is claimed. All 188 reviews now satisfy the unchanged source-bound guards. Earlier codebook versions and source/AI fields are preserved.

2. Cases

The following is Ryan's complete five-part individual case, with proposed interpretation clearly separated from his unconfirmed verdict. No other teammate case files were found in the current repository or fetched main. The team still needs legitimate distinct cases covering its main users and at least one difficult case; no teammate work is invented. The individual PDF remains separate from the team ZIP.

Ryan Yu case: Alex - should older-adult treadmill falls receive investigation priority?

Evidence draft for Ryan Yu's review, November 5, 2026. Not submission-ready. The reviewed production dataset and local numerical/UI checks are complete. Public deployment, genuine live screenshots and Ryan's final interpretation/verdict remain pending. No public use or personal conclusion is fabricated.

1. The person and the decision

Alex is a fictional CPSC epidemiology analyst, not someone interviewed. During investigation planning, Alex decides whether treadmill-related falls among adults 65 and older merit priority over other treadmill mechanisms for follow-up. A poor choice could divert limited investigative attention from a more consequential or informative pattern. The dashboard offers descriptive screening evidence, not a prescribed CPSC ranking rule.

2. Their real situation and alternatives

The CPSC NEISS Frequently Asked Questions describes screening surveillance data and follow-back investigation for additional circumstances. That supports the plausibility of Alex's role. It does not establish that CPSC uses this dashboard, that Alex is a real employee, or that burden alone sets priorities. Product association in an ED record does not prove that the product caused the injury.

The official NEISS Estimates Query Builder instructions document filters, records and optional coefficient-of-variation/confidence-interval outputs. That is one alternative, alongside a reconciled spreadsheet/script. Its broader product query is not automatically our narrative-screened, human-classified cohort. CPSC's Explanation of NEISS Estimates explains variability and estimate interpretation. Follow-back can add product/environment/medical circumstances absent from terse narratives. These tools complement our fast coordinated comparisons; they are not displaced by the app.

3. The app in use

Live link: PENDING authenticated hosting. The following are independently verified **local production results**, not observations of a live public app. No local screen is substituted for the required public screenshots. Capture each from the actual public URL, with active controls visible, and verify the same data hash before submission.

Screen and exact selections	Verified local production result; required public evidence
A: Fixed 2025; mechanism all; severity all; search blank	Headline 5,120 weighted visits / 109 records / 22.3% severe. Fall: 65 / 3114.4453 / 28.340164%. Belt contact: 9 / 330.4278 / 31.612080%. Equipment obstacle/landing: 8 / 444.8651 / 11.647823%. PENDING PUBLIC SCREENSHOT: comparison charts and supporting counts.
B: Mechanism Fall; severity all; search blank	65 records; 3114.4453 weight; 18 severe records / 882.6389 weight. Primary fracture weight 1092.5582, 35.080346% of Fall weight. Unknown outcomes zero; both sensitivity endpoints 28.340164%. PENDING PUBLIC SCREENSHOT: diagnoses and expanded sensitivity.
C: Fall; Severe; search blank, then belt	Severe selection: 18 / 882.6389 / 100%; all-outcome Fall baseline remains 28.340164%. Adding belt: zero records, zero weight, N/A; baseline retains Fall and belt. PENDING PUBLIC SCREENSHOT: labeled baseline and honest empty result.

Local checks: validation/independent_numbers.py independently reads CSV using Decimal; validation/verify_dashboard.py checks all builders, filter IDs, diagnoses, baselines, endpoints and CSV; validation/verify_app.py runs the actual configured app with no fixture loader. All PASS. CSV SHA-256 c97b9ef421000d5a6e476a5e1598e16d6c8fde9ad20b5cd32d83c626480e985f. Rounded headlines differ from underlying full-precision values by design. These commands and saved JSON reports are reproducible, but do not establish public availability or usability.

4. Reading the result for their decision

Does the measure fit? Fall has the greatest weighted burden of the six observed mutually exclusive categories: 3114.4453 of 5119.6280 weighted visits, supported by 65 of 109 sampled visits. Its 28.340164% severe share is above the all-cohort 22.302804%, but below Belt contact's 31.612080% from only nine records (two severe). Those descriptive contrasts are screening evidence; without survey-design uncertainty, they do not establish a stable difference. The app measures ED burden and recorded disposition, not preventability, investigation value or risk per user (kinds 3, 4 and 9 in Module 11).

The CPSC estimate guidance flags national estimates below 1,200, fewer than 20 cases, or CV above 33% as unstable. Every observed non-Fall mechanism is below the first two thresholds. Fall's 65 records / 3114.4453 pass those two screens, but its CV is uncomputed, so stability is not certified. Its severe subset (18 / 882.6389) is also small. These are national-estimate reporting cautions, not a significance test for differences in severe percentages.

Who is in the rows? One row is one ED visit, not one unique person. The full annual source has 410,201 records / 25 original columns; 188 candidates satisfy age 65-120, product 3277 and literal treadmill mention. Ryan's accepted injury-focused policy and genuine review retain 109 and exclude 79, including medical-only or incidental unrelated episodes. Other ages, non-ED injuries, missing mentions, other years and reviewed exclusions are absent (kinds 1, 2 and 5). The codebook's Fall is not every fall-involving injury: mounting, initiating belt and equipment obstacle/landing mechanisms are separated. The eight equipment cases do not establish power/use status. A broader all-fall question requires deliberate cross-category narrative analysis, not summing every belt or mounting case automatically.

What outside the data matters? Exposure to treadmill use, baseline health, medical triggers, investigation feasibility, preventability and new incident information could change priority. Neither the app nor these brief records supplies a causal comparison or usage denominator (kinds 7 and 8). Fractures contribute 35.080346% of Fall weight, which supports asking about clinical consequences in follow-up; it does not identify a design defect. A count is not statistical precision. The official uncertainty-capable tools and follow-back are essential complements.

What could Alex misread? The Fall + Severe headline is 100% because the selected denominator contains only severe visits; its explicit baseline remains 28.340164% (kind 6). The empty Fall + Severe + belt subset cannot show belt events never occur. All-cohort belt search actually returns three nonsevere records / 123.7879 weight. Outcome unknown weight is zero here, so the two endpoints coincide, but this does not eliminate sampling or eligibility uncertainty. Code 57 is the primary fracture bucket only, not every diagnosis. None of these figures measures injury probability per treadmill user.

5. The verdict: prove or break

Proposed conclusion for Ryan to evaluate, not his confirmed verdict: Fall's relatively large reviewed burden and 18 severe records justify placing it on a follow-up shortlist, while nine Belt contact records warrant caution about comparative severe percentages. Use the app to assemble questions and inspect evidence, not to declare a unique winner, a defect, or exposure-adjusted risk. Its coordinated baselines and exact reconciliation reduce comparison work relative to repeatedly reconciling a spreadsheet. The official query offers uncertainty; follow-back adds circumstances; a broader independent analysis can examine all fall-involving events. There is not yet evidence to prove this app is as good as it gets for Alex.

Possible break branch for Ryan's approval: make the absence of survey-design uncertainty visible beside the severe-outcome scatter, so a nine-record estimate is not mistaken for a precise ranking. Current supporting-count caveats help but do not estimate uncertainty. A change could add validated design-based intervals when the survey design supports them, or a clear adjacent statement that differences are descriptive. It should apply to every mechanism and persona; do not compute a naive binomial interval on weighted observations. This is a proposed nonduplicate request, not an implemented feature or team-approved change. The existing CR-01 through CR-08 already address baselines, counts, narratives, Undo, missingness, definitions and precision; their original descriptions/implementation evidence are preserved.

Ryan must inspect the genuine public screens, decide whether this limitation changes Alex's decision, and choose/substantiate prove or break. Criteria 6 and 7 remain incomplete without that live evidence and owned interpretation. The original claim and earlier draft reasoning are preserved in version history/backups; this memo updates facts without attributing an unconfirmed verdict.

3. Numbers

The guarded production workflow passed with **188 completed genuine reviews, 109 Include / 79 Exclude**, zero held cases and exactly 109 retained IDs. All retained weights match the original candidates as Decimal values. The full annual source has 410,201 records / 25 columns and dates January 1-December 31, 2025. Production CSV SHA-256: c97b9ef421000d5a6e476a5e1598e16d6c8fde9ad20b5cd32d83c626480e985f. Codebook version **1.2**. No candidate count is presented as an injury estimate.

Total T = **5119.6280 weighted ED visits**, 109 sampled visits. Severe dispositions 2/4/5/8 contribute **1141.8206** weight from **24** visits. S/T x 100 = **22.302804031855439496775937 63%** (headline 22.3%). Unknown disposition U = **0**, zero records. Both conditional sensitivity endpoints therefore equal 22.302804031855439496775937 63%; these are not confidence intervals.

Mechanism	Records	Weighted ED visits	Severe records	Severe weight	Severe %
Fall	65	3114.4453	18	882.6389	28.340164
Unknown/Unclassified	15	697.4813	3	102.9095	14.754446
Equipment/cord obstacle or landing falls	8	444.8651	1	51.8171	11.647823
Belt contact	9	330.4278	2	104.4551	31.612080
Repair/moving	8	277.7898	0	0	0.000000
Entering/exiting	4	254.6187	0	0	0.000000
Struck by equipment	0	0	0	0	N/A

All groups use all selected visit weight as the severe denominator. The codebook's Struck category is valid but absent in this cohort; the UI presents only observed mechanisms. Fall is not all fall-involving events: mounting, belt-initiated and equipment obstacle/landing mechanisms are distinct.

Primary diagnosis shares use one Diagnosis bucket per visit, divided by all weight in that mechanism; missing/unmapped buckets are preserved. In Fall, fractures (code 57) contribute **1092.5582 / 3114.4453 = 35.080346410322249037412858 08%**. The independent report enumerates every diagnosis bucket, not just this example.

All 14 independent scenarios pass: all, six observed mechanisms, each severity, literal belt, literal metacharacters, deliberate empty and Fall + Severe + belt. Unrestricted belt = 3 records / 123.7879 weight / zero severe. Fall + Severe + belt = zero records / zero weight / N/A, not evidence that belt injuries never occur. Severe-only all-cohort = 24 records / 1141.8206 weight / 100%; the comparison baseline remains 109 / 5119.6280 / 22.302804%, retaining other selections. Fall severe baseline remains 65 / 3114.4453 / 28.340164% versus 18 selected severe records / 882.6389 weight.

The independent csv/Decimal method shares no pandas aggregation with the application. It checks exact ID sets, counts, mechanism and diagnosis buckets; weights/percentages use max(1e-8 absolute, 1e-12 relative) tolerance for floating-point dashboard values. CSV contributions reconcile unrounded; the separate selected_total row is not another contribution. Mechanism values rounded separately sum to 5119, while the rounded total is 5120. This one-visit display difference is explained by rounding; no weights were changed. No unexplained discrepancy remains.

Evidence: validation/results/real_2025_results.json, real_2025_dashboard_check.json and real_2025_app_check.json. These establish reviewed-data, builder/export and actual configured local-app agreement, **not public-browser evidence**. Public CSV download and visible screens must still be compared with this same hash.

The earlier invented hand-worked fixture remains documented in validation/results/fixture_results.json and dashboard_fixture_results.json. It tests 120 weight / five records / 41.666667% severe and unknown endpoints 41.666667-75%; it is never substituted for real findings. Canonical integer-code normalization resolved an earlier fixture 53.0-versus-53 representation issue without changing metric definitions. Actual real-data discrepancies are fully resolved; public comparison remains a separate unfinished check.

4. Tests

Full repository suite: **180 passed, 0 failures, 0 errors, 0 skipped**, Windows/Python 3.13.15, JUnit 38.457 seconds. Results and module counts are in validation/results/test_summary.json. This fresh run includes the finalized runtime pair. Two additional taxonomy regressions beyond 178 protect rejection of the old Non-use label and separation of surface Fall from equipment relationship in baselines/exports. The missing-file AppTest now uses an explicitly absent temporary path, preserving that behavior test when production data is shipped; it was not deleted or weakened.

The original build had 87 tests; the team added 93. All eight CRs are protected by tests/test_mvp_changes.py. Preparation/source gates, duplicate/foreign/missing IDs, altered narratives, policy/candidate hashes, reviewer identities, deferred/missing approvals, formula attestations and unresolved taxonomy gaps are rejected. Invented end-to-end checks cover guarded export/finalization and no-overwrite publication; no production bypass flag exists. Software verifies consistency and identity, not whether a reviewer truly read a narrative; genuine approval rests on Ryan's explicit confirmations and adjudications.

Loader/provenance tests reject missing/edited/sample bundles, codebook and lookup mismatches. Filter/view tests protect literal membership, severity/diagnosis baselines, unknown denominators, empty states, reset/session-specific Undo and cache invalidation. Independent Decimal checks protect aggregate and CSV arithmetic; an incorrect-export injection is rejected. Colby's committed tests protect row-order invariance and uniform weight-scaling behavior. Git CRLF tests stage invented bytes and verify CSV/provenance/codebook hashes survive; production staged bytes are also compared directly. The first actual Ubuntu run caught one invented end-to-end fixture regenerating codebook text with Linux newlines, conflicting with the exact-byte guard. Copying canonical bytes fixed the fixture; the guard and production bytes were not changed. The subsequent full Windows suite passed again.

The actual configured local production AppTest uses no fixture injection and checks real metrics/plots/tables, baseline and matching narrative sections, zero selections, codebook and Reset/Undo. The actual Ubuntu CI run passed **180 tests, zero failures/errors/skips** in 12.00 seconds on Python 3.13.18, plus pinned dependency installation/checks, real independent numbers, dashboard filters/baselines/CSV and production AppTest. Its actual Streamlit health returned ok and root HTTP succeeded. The matching Windows-produced hashed bundle therefore loads on Linux. See deployment/CI_RESULTS.md and validation/results/linux_production_check.json. Local/CI checks do not establish public incognito access or provide live screenshots.

Release guards exclude raw downloads, private reviews/workbooks/audits, environments, caches and secret patterns. Final runtime CSV/provenance/codebook bytes are preserved exactly by Git attributes, and provenance publishes only a source basename. Source data, the original candidate/review CSV, inclusion-policy bytes and one-shot prompt remain preserved. The release guard still refuses a submission ZIP while README links and public/video evidence are pending. Account access, legitimate teammate cases, live screenshots, Ryan's verdict and recording remain genuine deliverable requirements; automated passes do not satisfy them.